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The role of antibodies in small fiber neuropathy: a review of currently available evidence

<https://doi.org/10.1515/revneuro-2024-0027>

Received February 16, 2024; accepted May 26, 2024;

published online June 13, 2024

Keywords: small fiber neuropathy (SFN); neuropathic pain; Covid19; FGFR3 antibodies; TS-HDS antibodies; neuronal antibodies

Abstract: Small fiber neuropathy (SFN) is a peripheral nerve condition affecting thin myelinated A δ and unmyelinated C-fibers, characterized by severe neuropathic pain and other sensory and autonomic symptoms. A variety of medical disorders can cause SFN; however, more than 50% of cases are idiopathic (iSFN). Some investigations suggest an autoimmune etiology, backed by evidence of the efficacy of IVIG and plasma exchange. Several studies suggest that autoantibodies directed against nervous system antigens may play a role in the development of neuropathic pain. For instance, patients with CASPR2 and LGI1 antibodies often complain of pain, and *in vitro* and *in vivo* studies support their pathogenicity. Other antibodies have been associated with SFN, including those against TS-HDS, FGFR3, and Plexin-D1, and new potential targets have been proposed. Finally, a few studies reported the onset of SFN after COVID-19 infection and vaccination, investigating the presence of potential antibody targets. Despite these overall findings, the pathogenic role has been demonstrated only for some autoantibodies, and the association with specific clinical phenotypes or response to immunotherapy remains to be clarified. The purpose of this review is to summarise known autoantibody targets involved in neuropathic pain, putative attractive autoantibody targets in iSFN patients, their potential as biomarkers of response to immunotherapy and their role in the development of iSFN.

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1 Introduction

Small fiber neuropathy (SFN) is a peripheral nerve disorder characterized by damage of thin myelinated A δ and unmyelinated C-fibers, the smaller diameter fibers involved in the perception of cold and warm sense, respectively, and the transmission of pain (Devigili et al. 2020; Hoeijmakers et al. 2012; Hovaguimian and Gibbons 2011). Moreover, A δ -fibers have a role in preganglionic sympathetic and parasympathetic functions, and C-fibers in postganglionic autonomic functions (Devigili et al. 2020; Hoeijmakers et al. 2012; Hovaguimian and Gibbons 2011).

SFN is relatively common, with a worldwide prevalence of 53–130 per 100,000 people and an incidence of 11.73 per 100,000 per year, probably largely underestimated due to the difficult diagnosis (Bitzi et al. 2021; Peters et al. 2013). For example, fibromyalgia patients often have undiagnosed SFN (Giannoccaro et al. 2014). SFN manifests as neuropathic pain (NeP)/allodynia, paresthesia, diminished sensitivity to pinprick and heat, and a range of autonomic symptoms such as dry eyes and mouth, constipation, urinary retention, sexual dysfunction, palpitations, and altered sweating (Devigili et al. 2020; Hoeijmakers et al. 2012; Hovaguimian and Gibbons 2011). Symptoms can be length-dependent (LD), mainly involving the toes with subsequent upward spread, or non-length-dependent (NLD), with involvement of the face, upper limbs, or trunk (Devigili et al. 2020; Hovaguimian and Gibbons 2011).

The diagnosis consists of a combination of clinical evaluation and diagnostic testing. Skin biopsy from the distal leg, with quantification of the linear density of intra-epidermal nerve fibers (IENFD), can be considered the gold standard for SFN diagnosis in association with clinical features of SFN involvement. An additional biopsy from the proximal thigh is helpful to inform about the predominant, length-dependent or non-length-dependent, SFN pattern through the measurement of the leg to thigh IENF density

ratio (Devigili et al. 2020; Lauria et al. 2010; Provitera et al. 2018). Other commonly used tests to support the diagnosis include quantitative sudomotor axon reflex testing (QSART), quantitative sensory testing (QST), sympathetic skin responses (SSR), or thermoregulatory sweat testing (TST), and most recently electrochemical skin conductance (ESC), corneal confocal microscopy (CCM), skin wrinkling, and laser evoked potentials (LEPs) (Devigili et al. 2020; Hovavimian and Gibbons 2011; Zeidman 2021). SFN can occur because of several medical conditions (Table 1), including diabetes, vitamin and micronutrient deficits, celiac disease, genetic mutations in sodium channel, infections, and amyloidosis (Peters et al. 2013; Zeidman 2021). However, over 50% of cases are idiopathic (iSFN) (De Greef et al. 2018). Some studies suggest a possible autoimmune etiology, supported by the observation of the efficacy of Intravenous Immunoglobulin (IVIG) and plasma exchange (PLEX) in some patients (Liu et al. 2018; Oaklander 2016). Nevertheless, a randomized controlled trial including 60 patients with painful iSFN showed that IVIG treatment had no significant effect on pain (Geerts et al. 2021), suggesting the need to identify possible biomarkers of immunotherapy responsiveness.

Recently, some clinical studies have revealed an association between SFN and the presence of different autoantibodies (Figure 1), allowing a proportion of cases previously considered idiopathic to be possibly reclassified as “immuno-mediated” SFN (Zeidman et al. 2022), although further validation and demonstration of pathogenicity is needed for most of them. This review aims to summarise

known autoantibody targets involved in neuropathic pain (Tables 2 and 3), potentially attractive autoantibody targets in iSFN patients (Tables 2 and 3) and in post-COVID-19/post Covid vaccine SFN, and comment on their function as possible biomarkers of potentially treatable neuropathic pain. Identifying new diagnostic markers would help in a more effective diagnosis and targeted patient treatment and, above all, point toward the fundamental pathogenicity studies for these purposes.

2 Search methods

Two Authors (MPG and LM) independently searched Pubmed and Scopus database using varying combinations of the following search terms, adding the Filter “English”: “small fiber neuropathy/painful neuropathy/neuropathies and autoimmune/autoimmunity/autoantibody/autoantibodies/neuronal antibodies/markers/biomarkers; covid-19/SARS-CoV2 and autoantibodies/autoantibody/autoimmunity and small fiber neuropathy”. Reference lists of included papers were manually browsed to identify any additional studies.

3 Autoantibody detection methods

Autoantibodies against neuronal targets can be detected through different methods, depending on the target type. Usually, methods that preserve conformational epitopes of the antigens, such as cell-based assay (CBA), flow cytometry, unfixed/postfixed tissue-based assay (TBA), or immunostaining on primary culture, are used to detect autoantibodies to surface antigens (Ricken et al. 2018; Van Coevorden-Hameete et al. 2016).

The CBA and flow cytometry use mammalian cells transiently transfected with plasmid to express the target antigen on their surface in its native form. CBAs are preferable for detecting neuronal surface Abs than other methods such as western blot (WB), radioimmunoassay (RIA) or peptide enzyme-linked immunosorbent assays (ELISAs), in which the proteins are not necessarily conformational (Waters et al. 2016).

TBA and the immunofluorescence on rat/mouse neurons in culture are often used as screening methods to detect antibodies against known and unknown neuronal and glial antigens. TBA consists of detecting the reactivity of antibodies present in patients’ serum or CSF against rat/mouse/primate brain section by immunohistochemistry (IHC) or immunofluorescence (IF). Specifically, for the detection of antibodies directed against targets of interest in neuropathies, serum reactivity is assessed on rodent DRG sections, spinal cords, teased nerve fibers, cultured rodent sensory neurons and

Table 1: Main SFN aetiologies.

–	<u>Idiopathic</u>
–	<u>Metabolic:</u>
–	Diabetes mellitus and defective glucose tolerance, Vitamin/micronutrient deficiencies (B12, B6, B1, E, and copper), Abnormal thyroid function;
–	<u>Immune-mediated:</u>
–	Systemic lupus erythematosus, Sjögren syndrome, Sarcoidosis, Rheumatoid arthritis, Celiac disease/gluten sensitivity, Connective tissue disease; Paraneoplastic; Monoclonal gammopathy;
–	<u>Toxic:</u>
–	Alcoholism, chemotherapy (bortezomib), B6 toxicity, Neurotoxic drugs, vaccine-associated, heavy metal toxicity;
–	<u>Infectious agents and post-infections:</u>
–	Lyme, human immunodeficiency virus (HIV), herpes simplex virus (HSV), varicella zoster virus (VZV), Cytomegalovirus (CMV), hepatitis-C virus (HCV), hepatitis-B virus (HBV), leprosy;
–	<u>Hereditary:</u>
–	Fabry’s disease, Mutation in sodium channels (SCN9A, SCN10A, SCN11A), Wilsons disease, Familial amyloidosis;
–	<u>Other:</u>
–	Fibromyalgia; Chronic pelvic and bladder pain; Complex regional pain syndrome type I; Ehlers-Danlos syndrome.

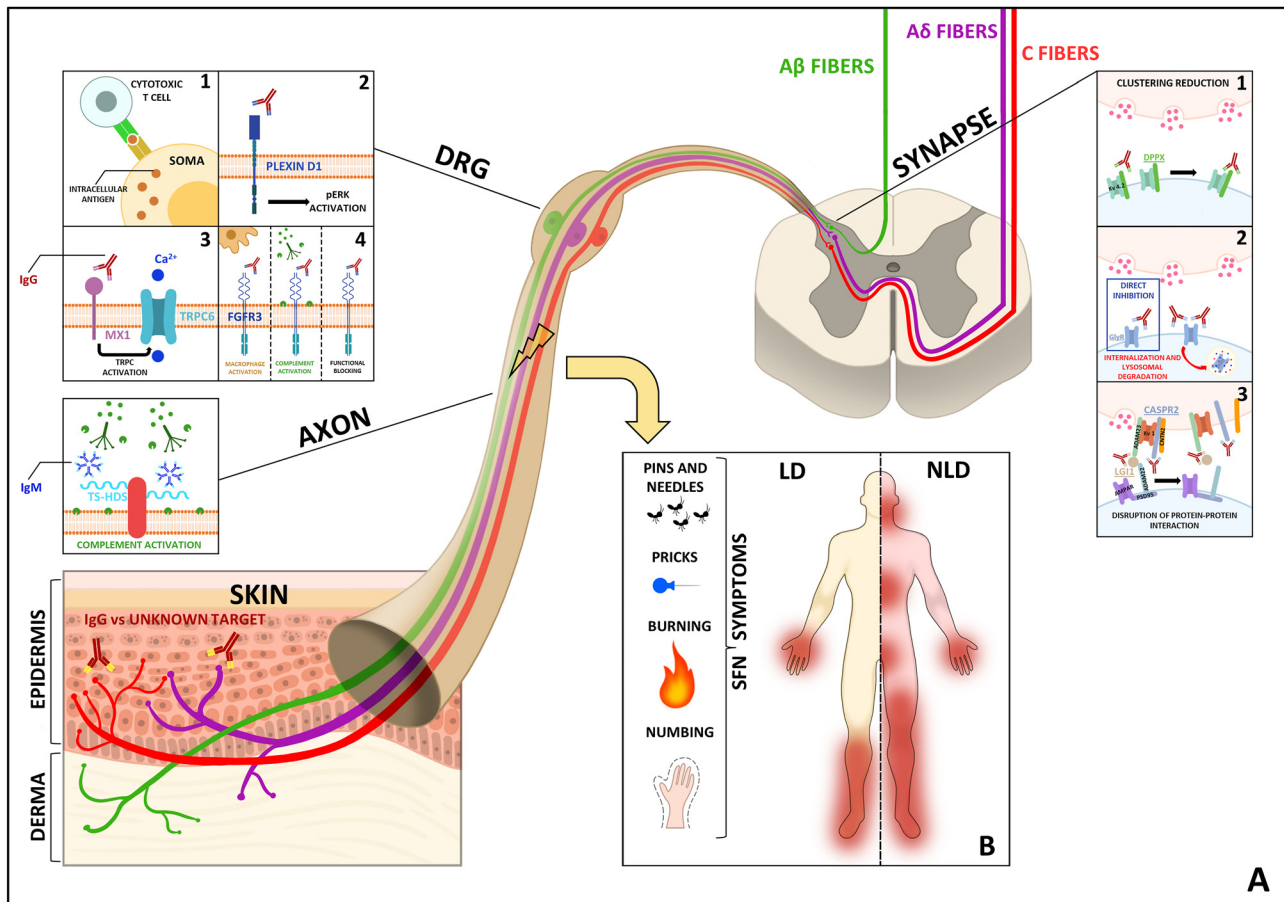


Figure 1: Schematic representation of the hypothetical role of autoantibodies in SFN pathogenesis: A. Autoantibodies could bind to unknown and known targets on sensory neurons at the level of the DRG, axons, synapses, and skin, causing fiber damage through various mechanisms, leading to neuropathic pain. Synapse: 1. DPPX IgG1 and IgG4 cause a decrease of DPPX clusters and Kv4.2 protein; 2. GlyR IgG1 and IgG3 can directly block the protein, or induce cross-linking resulting in internalization and degradation of the target; 3. LGI1 IgG4 (rarely IgG1) may disrupt the interaction with ADAM22 and consequent reduction of AMPAR, Kv1.1, and VGKC; possible complement activation; CASPR2 IgG4 – 1 (IgG4>>IgG1) induce the alteration of the interaction with associated molecules as well as internalization and complement activation. DRG: 1. Onconeural antibodies to intracellular neuronal proteins do not directly induce DRG neuronal damage which is mediated by T cells instead; 2. pERK activation by Plexin-D1 IgG1 – 2 binding; 3. Potential TRPC6 activation by IgG to MX1; 4. FGFR3 IgG1–3 (IgG1>>IgG3) could activate the complement cascade or induce antibody-dependent toxicity by macrophage, while IgG4 can cause functional blocking of receptor proteins. Axon: TS-HDS IgM could induce complement activation. Skin: IgG to unknown antigens could bind directly to intraepidermal nerve fibers. B. Patients show two sensory symptom distribution patterns: Length-dependent (LD) and non-length dependent (NLD) SFN. Sensory symptoms include altered sensations such as pins-and-needles, pricks, burning, and numbing.

human sensory neurons derived from induced pluripotent stem cells (iPSCs) (Clark et al. 2017; Doppler et al. 2016; Saporta and Shy 2017; Yuki et al. 2018).

Although ELISA has some limitations, including the loss of the protein native conformation and the unspecific binding of Abs to the solid phase that can increase the risk of false positive results, it is the most widely used technique for many targets. The inclusion of empty wells that have been treated the same as the target antigen-coated wells and optimization of the ELISA protocol by normalizing serum-specific background noise (Moritz et al. 2019; Waters et al. 2016), might limit the risk of unspecific binding.

Considering the various limitations of the techniques, the choice depends on the target, and it is essential to compare different methodologies and use a confirmation assay.

4 Immune disorders associated with small fiber involvement

4.1 Antibodies against neuronal antigens

Paraneoplastic neuropathic pain manifests in the context of sensory neuronopathies (SNN) due to the degeneration of

Table 2: List of the main antigenic targets discussed and associated with pain, focusing on SFN.

Target	Target localization	Technique of detection	Antibody iso-type and subclasse(s)	Antibody mechanism	Evidence of pathogenicity <i>in vitro</i> and/or <i>in vivo</i>	Main clinical phenotype(s)	Pain related syndrome	References
Onconeural (Hu, PCA2 (MAP1B), CRMP5/CV2 and amphiphysin)	Intracellular	TBA + dot-blot/CBA	IgG	Biomarkers; T cell mediated neuronal destruction	No evidence of direct pathogenic effect <i>in vivo</i> and <i>in vitro</i>	Rapidly progressive cerebellar degeneration, limbic encephalitis, encephalomyelitis, OMS, SPSD	SNN or sensory neuropathies	Zoccarato et al. 2021; Sommer et al. 2005; Tanaka et al. 1994; Hornigo and Lieberman 1994
CASPR2 and LGI1	Neuronal surface	TBA, CBA	IgG (IgG4>IgG1)	Altered interaction with partner proteins; possible internalization	Altered Kv1.2 expression and changes in neuronal excitability; increased sensitivity towards pain (CASPR2), memory impairment (LGI1, CASPR2)	Limbic encephalitis, FBDS (LGI1), cerebellar syndrome (CASPR2)	MoS, neuropathic pain syndromes, SFN, neuromyotonia	Irani et al. 2012; Klein et al. 2012; Laurencin et al. 2015; Dawes et al. 2018; Fernandes et al. 2019; Giannoccaro et al. 2019; Joubert et al. 2022; Paterson et al. 2018; Ramanathan et al. 2021; Gendre et al. 2024
DPPX	Neuronal surface	CBA, TBA	IgG1, IgG4	Reduction in DPPX clusters and Kv4.2 subtype potassium channels	Neuronal increased excitability and action potential rate	Limbic encephalitis/encephalopathy, gastrointestinal symptoms, PERM	Neuropathic pain	Bjerknes et al. 2022; Hara et al. 2017; Piegras et al. 2015; Tobin et al. 2014
GlyR	Neuronal surface	CBA	IgG1 and IgG3	GlyR internalization; possible complement activation; direct GlyR inhibition	Reduced glycinergic transmission; some motor deficit in mice; disturbed escape behavior in zebrafish	PERM, SPS	Neuropathic pain and allodynia	Soleimani et al. 2023; Carvajal-González et al. 2021; Crisp et al. 2019; Rauschenberger et al. 2020; Lynch 2009
TS-HDS	Cell surface	ELISA	IgM	Unknown; possible complement activation	N/A	/	Painful distal-predominant symmetric polyneuropathy and non-length dependent SFN	Pestronk et al. 2003; Pestronk et al. 2012; Kafaie et al. 2017; Levine et al. 2020; Zeidman and Kubicki 2021; Trevino and Novak 2021; Olsen et al. 2022; Chompoopong et al. 2023; Gibbons et al. 2023; Zeidman et al. 2022
FGFR3	Intracellular	ELISA, CBA (less sensitive than ELISA)	IgG (IgG1 > IgG4)	Unknown; possible complement activation or Ab-dependent toxicity by macrophage, functional	No effects <i>in vitro</i>	/	SNN, small fiber neuropathy/neuronopathy	Antoine et al. 2015; Dave and Smith 2018; Tholence et al. 2020; Levine et al. 2020;

Table 2: (continued)

Target	Target localization	Technique of detection	Antibody iso-type and subclasse(s)	Antibody mechanism	Evidence of pathogenicity <i>in vitro</i> and/or <i>in vivo</i>	Main clinical phenotype(s)	Pain related syndrome	References
Plexin D1	Cell surface	ELISA, TBA	IgG (IgG2 > IgG1)	Neuronal cytotoxicity without complement activation	Increased membrane permeability and neuronal swelling <i>in vitro</i> ; mechanical pain in mice; pERK activation	/	Neuropathic pain; trigeminal neuralgia, SFN	Trevino and Novak 2021; Zeidman and Kubicki 2021; Gibbons et al. 2023; Zeidman et al. 2022; Gendre et al. 2024 Zeidman et al. 2022; Fujii et al. 2018; Fujii et al. 2021

CBA, cell-based assay; ELISA, enzyme-linked immunosorbent assay; IFa, immunofluorescence assay; iSFN, idiopathic small fiber neuropathy; MoS, morvan Syndrome; OMS, opsoclonus–myoclonus syndrome; PERM, progressive encephalomyelitis with rigidity and myoclonus; RIA, radioimmunoassay; SFN, small fiber neuropathy; SN, sensory neuropathy; SNN, sensory neuronopathy; SPSPD, stiff-person spectrum disorder; TBA, tissue-based assay.

DRGs and their central and peripheral projections (Fargeot and Echaniz-Laguna 2021). In these cases, pain is associated with a NLD involvement of all sensory modalities, including proprioceptive ataxia, patchy and asymmetric sensory loss, and reduced or absent tendon reflexes. More rarely, pain is associated with sensory neuropathy or sensorimotor neuropathy (Table 2). Onconeural antibodies classically associated with SNN or sensory neuropathies include Type 1 antineuronal nuclear antibodies (Hu/ANNA1), Microtubule-associated protein 1B (MAP1B/PCA-2), Collapsin response mediator protein 5 (CRMP5/CV2) and amphiphysin (Zoccarato et al. 2021), mostly in relation to small-cell lung carcinoma and breast cancer. SFN has been seldom reported in patients with hematological malignancies, in association (Waheed et al. 2016) or not (Liu et al. 2015) with onconeural antibodies, suggesting that SFN is a rare manifestation of paraneoplastic neurological syndromes. On the other hand, studies specifically investigating the frequency of onconeural antibodies in patients with SFN are scarce. In a recent study, among 85 SFN cases (48 % iSFN), 3 % were defined as paraneoplastic, although the associated onconeural antibodies were not reported (Pál et al. 2020).

Better characterized is the possible SFN involvement in patients with antibodies (Abs) against what was previously defined as the voltage gated potassium channel (VGKC)-complex, and now identified as directed against Contactin-associated protein-2 (CASPR2) and Leucine-rich glioma-inactivated 1 (LGI1) (Table 2). Patients with neurological syndromes related to CASPR2- and LGI1-Abs present most often with autoimmune encephalitis, and less frequently with peripheral nerve hyperexcitability and/or dysautonomia (Gadoth et al. 2017; Irani et al. 2012, 2011, 2010). Some patients, mostly harboring antibodies against both CASPR2 and LGI1, present with Morvan syndrome, characterized by the association of encephalopathy, autonomic dysfunction, and peripheral hyperexcitability. Pain has been reported in different proportions of these patients, and some of them may present with relatively isolated neuropathic pain without other signs of PNS or CNS hyperexcitability (Gendre et al. 2021; Ramanathan et al. 2021). In one of the first case series of Morvan syndrome, pain was observed in up to 62 % of cases (Irani et al. 2012). In another study, including 316 VGKC-Ab positive patients, 159 (50 %) had pain, in isolation (28 %) or associated with other neurological symptoms (72 %), more often in association with CASPR2-Abs. Pain was acute/subacute in onset and neuropathic (58 %) or nociceptive (47 %), more often distal (49 %) or diffused to the whole body (27 %), whereas 12 % of cases reported head/facial pain. Pain was defined as burning (33 %), allodynic (28 %), tingling (21 %), lancinating (19 %), and pruritic (1 %); it required multiple medications and improved with immunotherapy

Table 3: Summary table of studies on the association of novel autoantibodies in neuropathy, with a focus on SFN.

References	Target	Antibody detection test	Disease	Study features	Clinical presentation	Treatment and outcome ^a	Main results
Pestronk et al. (2003)	TS-HDS	ELISA	SN	5 TS-HDS IgM positive patients (4 F; 1 M; mean age: 67.4 years) vs 226 disease controls	Predominant sensory axonal polyneuropathies, neuropathic pain	N/A	5 seropositive patients; TS-HDS IgM was monoclonal, selective, and limited to κ light chains.
Pestronk et al. (2012)	TS-HDS	ELISA	SN	71 patients with TS-HDS IgM vs 41 controls (sensory or sensory-motor polyneuropathies but no anti TS-HDS IgM)	Distal, predominantly sensory polyneuropathies. Pain or persistent paresthesias.	N/A	3.03 % (71/2,342), SN; ($n = 58$ with sensory or sensory-motor polyneuropathies; $n = 10$ with concurrent MAG IgM; $n = 3$ with asymmetric chronic motor neuropathies). Presence of complement deposition in skin and muscle biopsies
Kafaie et al. (2017)	TS-HDS	ELISA	iSFN	8 SFN patients (4 F; 4 M; mean age: 14.25 years)	Alteration of sensation for temperature, pinprick, and touch, neuropathic pain	N/A	62.5 % (5/8) seropositive cases Association between painful SFN and the presence of TS-HDS IgM
Olsen et al. (2022)	TS-HDS	ELISA	SFN	17 TS-HDS IgM positive patients (12 F; 5 M mean age 57.5 years)	Paresthesia and pain	TPE; 71 % (12/17) reporting symptomatic improvement/slowing progression of disease	100 % (17/17) TS-HDS IgM positive patients; 12 % (2/17) FGFR3-Ab positive patients;
Chompoopong et al. (2023)	TS-HDS	ELISA	Neuropathy (26 % SFN)	77 TS-HDS IgM positive patients (50 F; 27 M; median age: 48 years) vs 25 negative controls (evaluated for idiopathic neuropathy and TS-HDS IgM negative)	LD-peripheral neuropathy ($n = 20$, 48 %) and LD-SFN ($n = 11$, 26 %). 34 % no neuropathy.	42/77 patients received immunotherapies: IVIG ($n = 40$), IVMP ($n = 6$), OCS ($n = 13$), MMF ($n = 5$), RTX ($n = 4$), AZA ($n = 1$); post-immunotherapy improvement: 31 % (13/42) but no more frequent compared to seronegative patients;	77 TS-HDS IgM positive patients; limitations in the value of TS-HDS IgM as a biomarker.
Antoine et al. (2015)	FGFR3	Protein arrays, ELISA, CBA (less sensitive), immunocytochemical study on sensory neurons	SN (7.55 % SFN (8/106))	106 patients with SN vs 211 disease and HCs	Positive patients: NLD-SN, particularly with pain and trigeminal nerve involvement.	N/A	15.09 % positive patients (16/106); IgG specific for FGFR3, no other FGFRs; selective for cytoplasmic TK domain; Sensory neurons expressed FGFR3
Tholance et al. (2020)	FGFR3	ELISA	SN (20.19 % (86/426SFN))	426 SN patients vs 361 anti-FGFR3 negative patients	Mostly chronic, NLD-neuropathy, pain	N/A	15.26 % anti-FGFR3 positive patients (65/426); 11 SFN (12.8 %); anti-FGFR3 Abs primarily affects the DRG.
Levine et al. (2020)	TS-HDS and FGFR3	ELISA	SFN		Neuropathic pain, loss of sensation, or numbness	N/A	37 % (57/155) with TS-HDS IgM and 15 % (23/155) with

Table 3: (continued)

References	Target	Antibody detection test	Disease	Study features	Clinical presentation	Treatment and outcome ^a	Main results
				155 patients with cryptogenic SFN vs 71 controls (ALS)			FGFR3-Abs; mainly F; NLD-pathology associated with both Abs; TS-HDS Ab prevalence in patients with an acute onset
Trevino and Novak (2021)	TS-HDS and FGFR3	ELISA	SFN	322 SFN patients	Predominant autonomic symptoms	N/A	TS-HDS IgM positive patients: 28 % (90/322), FGFR3 positive patients: 17 % (54/322); mainly F; Possible association with autonomic symptoms.
Zeidman and Kubicki (2021)	TS-HDS and FGFR3	ELISA	iSFN	40 iSFN patients (95 % F); average age of onset 43.5 (SD 12.8)	Neuropathic pain, numbness, and/or tingling with loss of sensation	11 positive patients treated with IVIG or CS but 2/11 stopped treatment early for side effects; 7/8 reduction in pain and/or improvement in IENFD	55 % positive patients (22/40); 77 % TS-HDS IgM, 27 % FGFR3-Abs, 5 % double positive
Gibbons et al. (2023)	TS-HDS and FGFR3	ELISA	SFN	20 SFN patients with Abs to TS-HDS and/or FGFR3 but only 17 completed the study (9 placebo, 8 IVIG); (50 % F)	Neuropathic pain, loss of pain sensation	IVIG (8 patients) + placebo (9 patients); improvement but no differences between the two groups	80 % (16/20) anti TS-HDS Ab positive patients (8 receiving IVIG, 8 receiving placebo); 35 % (7/20) anti FGFR3 Ab positive patients (4 receiving IVIG, 3 receiving placebo)
Zeidman et al. (2022)	TS-HDS, FGFR3 and plexin D1	N/A	iSFN	54 patients with iSFN (40 F, 14 M) = 24 seropositive vs 30 seronegative	Neuropathic pain, dysautonomia, itching, muscle cramps, and truncal, hand, and upper extremity symptoms, and less lower extremity symptoms and numbness	8 Ab-positive patients received IVIG, 6 patients received IVIG for at least six months; in patients who completed IVIG (6 patients for at least 6 months treatment), exams and questionnaires improved, and mean IENFD increased by 297 %	44.4 % (24/54) seropositive patients (62.5 % anti TS-HDS Ab, 29.2 % anti FGFR3 Ab, 20.8 % anti plexin D1 Ab);
Fujii et al. (2018)	Plexin D1	IFA	NeP	160 patients = 110 NeP patients (73 F, 37 M) vs 50 controls (27 F, 23 M)	Burning sensation (54.5 %), tingling sensation (54.5 %), and thermal hyperalgesia (45.5 %)	7 positive patients received immunotherapies with NeP improvement	10 % (11/110) seropositive NeP patients; Plexin D1-IgG affected the morphology of mouse DRG neurons <i>in vitro</i> ;
Fujii et al. (2021)	Plexin D1	ELISA (se. 75 %, sp. 100 %); TBA for confirmation (coincidence between tests 96.6 %)	SFN	118 patients = 63 patients with putative SFN (32 F, 31 M) vs 55 controls (37 F, 18 M)	Chronic persistent pricking or burning pain in a length-dependent distribution	N/A	12.7 % (8/63) seropositive patients; Plexin D1-IgG mice passive transfer induced mechanical pain;

Table 3: (continued)

References	Target	Antibody detection test	Disease	Study features	Clinical presentation	Treatment and outcome ^a	Main results
Yuki et al. (2018)	Nociceptive neurons	Serum reactivity on skin sections of mouse plantar pad, DRG and lumbar spinal cord.	SFN	3 Chinese patients with acute-onset SFN (3:M; mean age 27.33)	Hyperesthesia and brush allodynia in a glove-and-sock distribution, but also autonomic symptoms such as palpitations and severe constipation	2 patients received IVIG, 1 CS. Improvement with immunotherapy	IgG strongly bound to murine small nerve fibers in acute phase; staining disappeared during the convalescent phase. Serum transfer to a murine nociceptive model induced transient alteration in thermal pain responses.
Chan et al. (2022)	MX1, DBNL, KRT8	Proteomics (sengemics immunome protein array) and bioinformatics analysis	iSFN	Main cohort: 58 SFN patients (27:F;31:M; mean age:50.0 (±13.6) vs 20 HCs) – Validation cohort: 33 SFN patients (26:F;7:M; mean age:46.3 ± 12.5 years vs 20 HCs)	Most patients presented with positive symptoms (51.7 %) and in a LD pattern (51.7 %). Autonomic symptoms were prevalent in most patients (55.2 %).	N/A	Identification of autoantibodies to MX1, DBNL, and KRT8 in iSFN. Subgroup analysis into iSFN and secondary SFN suggests MX1 may allow diagnostic subtyping of iSFN patients.
Gendreau et al. (2024)	FGFR3, CASPR2	Serum reactivity on mouse sciatic nerve fiber or DRG by IFA	Acute-onset (<4-week) SFN (AOSFN)	42 patients = 20 patients (60 % women, median age [interquartile range] 44.2 years [35.7–56.2]) vs 10 HCs and 12 disease controls.	Pain (85 %); autonomic involvement (60 %); paresthesia (70 %); main sensory symptoms: Pinprick (90 %), responses to heat and cold (90 %). The clinical pattern was predominantly NLD (85 %).	6/20 patients treated with OCS (n = 3), outcome: complete or partial recovery; IVIG (n = 2), outcome: no efficacy; or TPE (n = 1), patient with anti-CASPR2 antibodies. Outcome: any efficacy.	22.2 % (4/18) FGFR3-Ab positive patients; 5 % (1/20) CASPR2-Ab positive patient; 70 % (14/20) mouse sciatic nerve fiber or DRG immunostaining; main clinical pattern: associated with acute NLD symmetric painful neuropathy with a monophasic or relapsing chronic course; Eight patients (32 %) had distal sensory impairment predominantly involving pinprick, temperature, and to a lesser degree light touch related to SFN
Dabby et al. (2000)	Sulfatide	ELISA	Neuropathy	25 of 450 patients examined (5.5 %) with significantly high Ab titers (0.25600)	Antibodies associated with several subtypes of peripheral neuropathy. Predominantly sensory or sensorimotor axonal neuropathies, with the sensory component either small fiber or mixed fiber type.	NA	
Moritz et al. (2023)	AGO1	ELISA, CBA (less sensitive)	SNN	823 patients = 132 SNN, 301 non-SNN neuropathies (including 116 CIDP, 80 SFN, and 105 ONPs), 274 AIDs vs 116 HCs	Paresthesia and dysesthesia in the face and global areflexia; abnormal ENMG	SNN patients (22/42 first line treatment; 3/42 s line and 17/42 both line of treatment); Improved outcome in patients AGO1 Ab positive, only for the use of IVIG.	Positive patients for AGO1 Abs observed in 3.8 % (3/80) SFN cases; Ab prevalent in patients with SNN

^aTreatment and outcome refer to positive patients or, if it refers to other classes of patients, it is specified.

(Klein et al. 2012). Nerve conduction studies, performed in 59 % of the pain cohort, were normal in most cases or showed mild sensory alterations. This finding, together with the presence of heat-pain hyperalgesia (C fiber allodynia) at the QST in more than one third of the tested cases with pain, suggested a predominant involvement of SFN, which however was not confirmed by skin biopsy (Klein et al. 2012). Nonetheless, Laurencin et al. (2015) confirmed the presence of reduced IEFND at the skin biopsy in three out of four patients with Morvan syndrome and pain. The fourth patient showed normal IEFND but abnormal LEPs, suggesting a functional SFN disorder possibly related to CASPR2-Abs-mediated increase in sensory neuron excitability. Indeed, CASPR2-Abs have been demonstrated to be pathogenic both *in vitro* and *in vivo* (Dawes et al. 2018; Fernandes et al. 2019; Giannoccaro et al. 2019; Joubert et al. 2022; Patterson et al. 2018). Peripheral passive transfer of CASPR2-IgG induced a significant mechanical pain related hypersensitivity in mice (Dawes et al. 2018). Moreover, DRG neurons incubated with patients' CASPR2-Abs showed a loss of Kv1 channel membrane expression and hyperexcitability. CASPR2-IgG injected mice displayed CASPR2-IgGs in the DRG and decreased expression of CASPR2 and Kv1 at the juxtaparanode without loss of epidermal nerve fibers (Dawes et al. 2018).

A recent study investigated the frequency, features and possible mechanisms of pain in 147 patients with CASPR2/LGI1-Abs (Ramanathan et al. 2021). This study confirmed the prevalent association with neuropathic pain of CASPR2-compared to LGI1-Abs. Both the LGI1- and CASPR2-Abs cohorts had predominant length-dependent distributions of pain, with truncal involvement exclusive to LGI1 antibody positive cases. Pain response to analgesic treatment was similar between the two groups, however pain improvement after immunotherapy was more frequent in patients with LGI1-Abs compared to those with CASPR2-Abs. Skin biopsy, performed in four cases, showed reduced IEFND in three cases (two LGI1- and one CASPR2-Abs positive). Interestingly, 38 % of CASPR2-Abs bound to live cultured unmyelinated human iPSC-derived sensory neuron versus none LGI1-Abs and 63 % of CASPR2-Abs bound to live murine DRG cultures compared to 7 % of LGI1-Abs independently from the presence of pain (Ramanathan et al. 2021). These results confirm, on one hand, the SFN involvement in patients with CASPR2/LGI1 autoimmunity, and on the other suggest the involvement of other modulating factors in patients with pain (i.e. accessibility of the target in peripheral nerve and sensory neurons, central mechanisms). Indeed, are these results consistent with effects of CASPR2/LGI1 antibodies on potassium channel function or could they be acting centrally through another mechanism? Further studies should better characterize these aspects.

Neuropathic pain has been rarely reported in patients with other antibodies against neuronal surface antigens (Table 2). For instance, antibodies targeting dipeptidyl-peptidase-like protein 6 (DPPX), a subunit of Kv 4.2 potassium channels, typically associated with encephalitis and gastrointestinal symptoms, have been reported in a patient presenting with pain and allodynia before the onset of cognitive impairment, although small fiber involvement was not investigated (Bjerknes et al. 2022). Nevertheless, among the cases of DPPX encephalitis reported to date, 8 % showed sensory symptoms including allodynia and pruritus, suggestive of unmyelinated C-fibers and thinly myelinated Aδ nerve fibers involvement (Bjerknes et al. 2022). DPPX-Abs induce neuronal hyperexcitability *in vitro* and *in vivo* (Hara et al. 2017; Piepgras et al. 2015; Tobin et al. 2014); however, their role in pain needs further investigation.

Finally, a recent study reported severe neuropathic pain and allodynia in a patient with brainstem and spinal syndrome due to glycine receptor antibodies (GlyR-Abs) (Soleimani et al. 2023). These antibodies are mainly associated with progressive encephalomyelitis with rigidity and myoclonus (PERM) and stiff person syndrome (SPS) (Carvajal-González et al. 2014). GlyR-Abs have been shown to be pathogenic *in vivo* and *in vitro*, acting through a reduction of the inhibitory glycine currents with consequent hyperexcitability (Carvajal-González et al. 2021; Crisp et al. 2019; Rauschenberger et al. 2020). Since GlyR are expressed in the DRGs and in the dorsal horn of the spinal cord and participate in the inhibition of nociceptive transmission (Lynch 2009), it is tempting to speculate about a possible role of these antibodies in patients with neuropathic pain. Therefore, their role in SFN warrants further investigation.

5 Antibodies associated with small fiber neuropathies

5.1 Antibodies against TS-HDS and FGFR3

Trisulfated heparin disaccharide (TS-HDS) is the most abundant disaccharide component of heparin sulfate glycosaminoglycans expressed at peripheral nerve sites. It can link extracellular proteins involved in different functions, such as the fibroblast growth factor family and their receptors that play a role in neuronal maintenance and repair (Grothe et al. 2001).

Antibodies targeting TS-HDS have been previously associated with sensory and sensory-motor polyneuropathies (Pestronk et al. 2012, 2003) (Tables 2 and 3). Pestronk et al.

(2003) firstly described five patients with painful, predominantly sensory axonal neuropathy and high titers of serum immunoglobulin M (IgM) binding to the trisulfated disaccharide IdoA2S-GlcNS-6S (TS-HDS); TS-HDS IgM was monoclonal and limited to κ light chain. Nerve biopsy documented an axonal neuropathy, with predominant loss of unmyelinated axons. Subsequently (Pestronk et al. 2012), the same Authors investigated the clinical features of a cohort of 58 TS-HDS IgM-positive patients with sensory or sensory-motor polyneuropathies. All patients had a distal-predominant, symmetric sensory loss involving the legs and, in 48 % of cases, the arms; 66 % of patients had persistent paresthesia or pain, involving both the legs and hands in 40 %, and more than half of cases showed a reduction of vibration sense. Given the prominent pain with loss of unmyelinated axon, further studies investigated the presence of TS-HDS antibodies in SFN patients. In a small cohort of pediatric patients with SFN, 62 % (5/8) had TS-HDS antibodies. All patients displayed a low IENFD at skin biopsy, clinically accompanied by a symmetrical alteration in temperature, pinprick, and touch perception in the legs (Kafaie et al. 2017). In a larger cohort, including 155 patients with iSFN 37 % had TS-HDS IgM compared to 11 % of ALS control patients. Antibodies were significantly more frequent in females, and were associated with NLD SFN pattern and an acute onset (Levine et al. 2020). Despite the initial association with sensory SFN, Trevino and Novak (2021) suggested the presence of autonomic dysfunction in seropositive TS-HDS-Abs SFN patients. In a cohort of 322 patients, 28 % had increased TS-HDS-Abs and manifested autonomic symptoms, including orthostatic, vasomotor, sudomotor, gastrointestinal, sexual in men, urinary, and pain.

Another attractive target is Fibroblast Growth Factor Receptor 3 (FGFR3) (Tables 2 and 3). FGFR3 is a cell surface protein belonging to the FGFRs family, four receptors expressed in the peripheral and central nervous systems. FGFR3 is involved in different functions, including nerve regeneration and axonal development (Trevino and Novak 2021). In adult rat DRGs, FGFR3 was expressed in small and large neurons and satellite glial cells (Antoine et al. 2015). Antoine et al. (2015) selected a cohort of 106 sensory neuropathies to search for specific antibodies using human ProtoArray V.4.2, confirmed by ELISA. Overall, 16/106 patients had FGFR3 IgG (11/38 with an autoimmune context, 5/46 with idiopathic neuropathy) and three had a prevalent or isolated SFN. Onset tended to be progressive, with more frequent face involvement. Nerve biopsy in six patients showed moderate to severe myelinated fibers loss. In a subsequent study (Tholance et al. 2020), the clinical features of sensory neuropathies associated with FGFR3-Abs were further investigated. In a large cohort of 426 patients with SN

(including 84/426 patients classified as SFN), 15.3 % (65/426) had FGFR3-Abs. Among them, 11 (11/86, 12.8 %) were classified as SFN. Antibodies were more common in females and in patients with a chronic disease course. FGFR3 antibodies were associated with other autoimmune diseases in 34 % of cases, and, overall, were the only markers of an autoimmune reaction in 66 % of cases. Nevertheless, immune treatment was not associated with improvement but with deterioration. The neuropathy had a non-length dependent distribution in about 89 % of cases suggesting that the DRG were the main target of the disorder.

In another study, 21 % (3/14) of FGFR3-Ab positive cases had a NLD SFN (Nagarajan et al. 2021). Further studies assessed the association of FGFR3 antibodies and SFN. Levine et al. (2020) found that 15 % (23/155) of iSFN patients was positive for FGFR3-Abs. Seropositive patients were mainly female and had a NLD clinical pattern. In a cohort of 322 SFN patients, 16.8 % had FGFR3-Abs. Patients complained of generalized autonomic dysfunction, although dysautonomic symptoms were present in similar proportions among seronegative and seropositive cases. However, FGFR3-Abs were more common among patients with mixed, somatic and autonomic, SFN, and combined functional and morphological SFN (Trevino and Novak 2021).

5.1.1 Evidence of pathogenicity of TS-HDS and FGFR3 antibodies

Despite the clinical associations reported so far, the relevance of TS-HDS and FGFR3 antibodies remains unclear. No *in vivo* or *in vitro* studies clearly demonstrated the pathogenicity of TS-HDS and FGFR3 Abs.

It is well known that IgG1 and IgM can efficiently activate the complement cascade. Muscle and nerve biopsies from patients with TS-HDS IgM showed abnormal C5b9 complement deposits on endoneurial and endomysial capillaries, suggesting a possible complement activation (Pestronk et al. 2012). However, there was no evidence of direct binding of serum IgM to axons or myelin nor other evidence of complement consumption was investigated.

FGFR3 antibodies were shown to be specific for this receptor and did not target other FGFRs (Antoine et al. 2015). In terms of IgG subtypes, IgG1 was the most frequent, in some cases associated with IgG3, which can activate the complement cascade or induce antibody-dependent toxicity by macrophage, whereas 30 % of cases had IgG4, which can cause functional blocking of receptor proteins (Antoine et al. 2015; Tholance et al. 2020). Nevertheless, these potential antibody effects were not demonstrated *in vivo* or *in vitro*. Moreover, the antibodies target an intracellular protein domain, the cytoplasmic tyrosine kinase domain (TKD),

composed of two subdomains, TRK1 and TRK2. A further study showed that the antibodies mostly target epitopes located in the juxtamembrane domain (JMD) and TK domain of the protein, that are crucial for its functioning (Tholance et al. 2021). Patients with antibodies against TKD exhibited a more severe clinical and electrophysiological impairment, supporting a pathogenic role of the antibodies (Tholance et al. 2021). FGFR3 is expressed in the nucleus and cytoplasm of DRG neurons. However, IgGs reacted only to the cytoplasmic target in immunostaining experiments on sensory neurons (Antoine et al. 2015). Therefore, it remains unclear whether and how antibodies directed against an intracellular epitope might affect the protein function.

Although these antibodies have been mostly reported in patients with sensory neuropathies, even their association with a defined clinical phenotype has been recently questioned. Chompoopong et al. (2023) demonstrated that the phenotypes of TS-HDS seropositive patients are heterogeneous, not limited to non-length dependent sensory predominant neuropathy. In their cohort of 77 patients with TS-HDS antibodies, one-third did not have any objective evidence of neuropathy and 12 % had other known causes of neuropathy. Similar results were reported in another study, which found that 40 % of seropositive patients had no evidence of neuropathy (Gibbons et al. 2023). In the study by Nagarajan et al. (2021), 25 patients with sensory or sensorimotor polyneuropathy with suspected autoimmune etiology had FGFR3-IgG. Four patients in this cohort evidenced a neuropathy unrelated to FGFR3-Abs. A critical limitation could concern the sensitivity and specificity of indirect ELISA. Moritz et al. (2019) strongly recommend normalizing results on serum-specific background noise when working with serum, as it reduces the risk of false positive and false negative results. The criticality of the assay was highlighted in another small study, including seven patients with FGFR3-IgG (Samara et al. 2018). In three of the seven patients, repeat antibody testing a few months later without any intervening immunotherapy showed no FGFR3-Abs in two and a significant titer reduction in the last patient. The Authors recommend practicing caution in interpreting these antibody titers in patients with neuropathy, suggesting a repeat of the detection assay, which has high variability.

Even if TS-HDS and FGFR3 antibodies could be not pathogenic, they could serve as biomarkers of an autoimmune SFN and therefore as predictors of response to immunotherapy. In a retrospective study, iSFN patients with TS-HDS and FGFR3-Abs showed some improvements after IVIG treatment, especially in skin innervation density and reduction in VAS pain scores (Zeidman and Kubicki 2021). A subsequent retrospective study including 54 cases of iSFN

revealed that 44.4 % of patients had auto-antibodies (62.5 % TS-HDS, 29.2 % FGFR-3, 20.8 % Plexin D1) (Zeidman et al. 2022). IgG-positive patients tended to have a higher SFN-SL score, an abnormal QSART, and previous misdiagnosis. Following IVIG treatment, there was notable enhancement in the test results, with a remarkable 297 % increase observed in the mean IENFD (Zeidman et al. 2022). In a retrospective cohort of 17 subjects with TS-HDS-Abs, 71 % reported symptomatic improvement or slowed disease progression after plasma exchange treatment (Olsen et al. 2022), although it is unclear how the outcome was measured. Despite these reports, a double-blind placebo-controlled study of IVIG in a cohort of suspected immune-mediated SFN associated with TS-HDS- or FGFR3-Abs, detected no improvements in nerve density, pain, or physical examination findings (Gibbons et al. 2023), questioning their role as biomarkers of immune responsiveness in iSFN patients.

5.2 Plexin D1 antibodies

Plexin D1 is a transmembrane receptor belonging to the plexin family, which consists of four classes (A, B, C and D) (Hota and Buck 2012; Zhang et al. 2021). Plexins interact with semaphorins, extracellular signaling proteins, and within the semaphorin family, the main interactor of Plexin D1 is the SEMA3E protein (Negishi et al. 2005; Zhang et al. 2021). SEMA-Plexin D1 signaling plays an important role in nervous, cardiovascular and immune system development, and it also plays a role in cancer biology. During nervous system development, in mouse embryogenesis, Plexin D1 is expressed within the central nervous system (Van Der Zwaag et al. 2002). SEMA3E activates Plexin D1, inducing cell repulsion and playing a role in neuronal wiring, axonal growth, and synapse formation (Chan and Wilder-Smith 2016; Zeidman 2021).

In recent years, a number of clinical studies have highlighted Plexin D1-IgG as a potential biomarker associated with NeP and SFN (Tables 1 and 2). A retrospective study performed on a cohort of 110 NeP patients and 50 controls, including 20 healthy subjects (HC) and 30 subjects with neurodegenerative diseases, identified Plexin D1-IgG through their binding to mouse DRG sections with a higher frequency in NeP patients than in non-NeP subjects (10 % vs 0 %; $p < 0.05$) (Fujii et al. 2018). Plexin D1-IgG exhibited a specific affinity for unmyelinated pain-responsive neurons situated in the DRG, where the blood brain barrier is not present, allowing unrestricted access for autoantibodies (Fujii et al. 2021). Antibodies were more frequent in females and were associated with younger age at onset. Patients presented with burning pain, thermal hyperalgesia and peripheral vascular dysfunction symptoms. Underlying neurological conditions

associated with Plexin D1 antibodies included atopic myelitis, NMOSD, MS, neurosarcoidosis and erythromelalgia. Immunotherapies ameliorated NeP in all treated cases (Fujii et al. 2018).

In a retrospective study carried out by the same authors, sera from 63 patients with putative SFN were analyzed. Plexin D1-IgG was positive in 8 of 63 (12.7 %) of all patients with SFN by ELISA, including 6 of 38 (15.8 %) patients with iSFN, 2 of 25 (8.0 %) patients with secondary SFN (sSFN), both of whom had diabetes, and 2 of 55 (3.6 %) HCs. Through a novel ELISA assay, Plexin D1-IgG was detected with a sensitivity of 75 % and specificity of 100 %, confirmed by the TBA assay (Fujii et al. 2021). In this cohort no female predominance was observed; the presence of Plexin D1-IgG was associated with late-middle age onset, chronic disease course (100 %), and burning feet (85.7 %). Disease duration at the time of serum sampling was significantly longer in SFN patients with Plexin D1-IgG than in those without and seropositive patients showed a higher frequency of pricking pain compared to seronegative ones (62.5 % vs 21.8 %, $p = 0.0278$).

Plexin D1-IgG have shown pathogenic potential in both *in vitro* and *in vivo* studies. Sera from patients positive for Plexin D1-IgG (belonging to the IgG2 or IgG1 class) directly increased membrane permeability and caused cellular swelling in cultured DRG neurons, all without triggering complement activation (Fujii et al. 2018). Among the various functions of sema-plexin signaling there is the regulation of cytoskeleton dynamics (Gay et al. 2011), so the authors propose a mechanism of cell death in DRG neurons mediated by cytoskeleton activation (Fujii et al. 2018). The same Authors also point out that common comorbidities of SFN include mainly allergic diseases, which increase the production of autoantibodies (Fujii et al. 2019). In mice passive transfer experiments, Plexin D1-IgG (purified from sera of seropositive patients) caused mechanical and thermal hypersensitivity, indicating a potential pathogenic effect *in vivo* (Fujii et al. 2021). The study suggests that Plexin D1-Abs are pathogenic but with low prevalence and holds promise as a potential biomarker for implementing immunotherapy in the context of SFN (Fujii et al. 2021). Zeidman et al. (2022) identified Plexin-D1 antibodies in 9 % of iSFN patients (5/54), confirming the relative low frequency of these antibodies.

The *in vitro* and *in vivo* pathogenic potential shown by Plexin D1-IgG leaves room for the possibility of undertaking therapies via IVIG and plasma exchange in order to block or remove circulating antibodies and relieve pain and other symptoms in patients with SFN (Zeidman et al. 2022). However, further prospective clinical trials are needed to confirm the efficacy of immunotherapy in Plexin D1-IgG-positive SFN patients.

5.3 Other antibodies in SFN

Other possible targets have been suggested, mostly in single reports (Table 3).

Sulfatide is a major glycosphingolipid in myelin and autoantibodies against this target have been associated with different forms of peripheral neuropathy, including demyelinating and axonal sensory-motor neuropathies. Dabby et al. (2000) have reported on the clinical, electrophysiological and pathological data of 25 patients with significantly elevated sulfatide antibodies. By reviewing the data, they identified four subgroups; the first group included eight patients with SFN which represented the largest group in their population accounting for 32 % of patients. Patients were characterized by distal sensory impairment predominantly involving pinprick, temperature, and, to a lesser degree, light touch. Quattrini et al. (1992) screened the serum from 200 patients with neuropathy and disease and healthy controls by ELISA. Five patients, with various clinical phenotypes showed anti-sulfatide IgM, including two with a clinical syndrome consistent with SFN, with normal electrophysiological or nerve biopsy studies, suggesting that increased titers of sulfatide Abs are found in patients with sensory impairment but are not restricted to a particular neurological syndrome or type of neuropathy.

The transient presence of neuronal antibodies was reported in three Chinese patients with acute onset SFN (AOSFN) presenting with severe pain in the extremities weeks after a previous infectious disease (Yuki et al. 2018). The patients manifested various symptoms such as hyperesthesia and brush allodynia in a glove-and-sock distribution, but also autonomic symptoms such as palpitations and severe constipation. Skin biopsy, performed in all patients, showed the absence of intraepidermal nerve fibers. To determine whether the patients had autoantibodies directed against nociceptive neurons, sera from the three patients were incubated on skin sections of mouse plantar pad, DRG and lumbar spinal cord. The acute phase sera, but not the convalescent phase ones, stained small nerve fibers in the dermis of foot pad skin and colocalized with nerve marker protein gene product 9.5 (PGP 9.5). Patients' IgG immunostained small neurons in the DRG, notably those with a higher density of voltage-gated sodium channels, and the dorsal horn of the lumbar spinal cord. Incubation of live DRGs in cultures with the sera showed that the acute phase patients' IgGs reacted against surface antigens expressed on the neuron cell body and axon of nociceptive neurons. Finally, it was seen that passive transfer of acute phase serum from two patients but not convalescent or control serum, to a murine recipient induced a transient thermal

hypersensitivity, supporting the idea that pain is mediated by immune-mediated mechanisms in AOSFN (Yuki et al. 2018).

A more recent study investigated the clinical features and associated autoantibody profile in a cohort of 20 patients with AOSFN (Gendre et al. 2024). Overall, one patient had CASPR2-Abs and 4/18 concomitant FGFR3-Abs. Sera IgGs from 14 patients (70 %) showed either mouse sciatic nerve fiber or DRG immunostaining. The sera from three patients showed paranodal immunostaining of Schwann cells or axons. One serum had antibodies that caused juxtaparanode immunostaining, consistent with CASPR2-Abs, but also unmyelinated fiber immunostaining. The sera from five patients bound to unmyelinated fiber whereas five contained antibodies against DRG. Altogether these results support the presence of antibodies to neuronal proteins in most patients with AOSFN, although their targets and pathogenic potential remain unknown.

Using a new technology called Sengenics Immunome Protein Array 20, a research group has recently identified novel autoantibodies associated with iSFN (Chan et al. 2022). This high-throughput technology uses a complete and functional human protein array to detect autoantibodies. The study recruited two cohorts of patients, including a main one with 58 adults with SFN and 20 HC matched for age and sex and a validation cohort of 36 patients and the same HCs.

Proteomic data showed the presence of 11 significant autoantibodies when comparing serum samples of SFN and HC in the main cohort. Among the 11 proteins, nine showed significant reproducible differences between SFNs and HCs, including Interferon-induced GTP-binding protein (MX1), Drebrin-like protein (DBNL) and keratin type II cytoskeletal 8 (KRT8).

MX1 is an interferon-induced GTP-binding protein that affects the activity of TRPC6, a transmembrane channel that allows calcium to pass through cells. DBNL is a protein involved in receptor-mediated endocytosis and cytoskeleton reorganization. KRT8 is a protein found in striated muscles, and its gene is downregulated in patients with chronic inflammatory demyelinating neuropathy (CIDP). A subanalysis comparing idiopathic and secondary iSFN showed that MX1 antibodies maintained significantly higher levels in the group with iSFN, suggesting that they may be an important biomarker for this condition. However, these data need confirmation in other cohorts.

Finally, a recent study investigated Argonaute (AGO) antibodies in patients with neuropathies (Moritz et al. 2023). Antibodies against AGO1 and AGO2 were initially identified in patients with non-paraneoplastic autoimmune central and peripheral neurological disorders including sensory neuronopathy and limbic encephalitis (Do et al. 2021). Subsequently, Moritz et al. (2023) investigated by ELISA the

frequency of AGO1-Abs in patients with sensory neuronopathy (SNN). AGO1-Abs, mainly IgG1, were more common in patients with SNN (17/132, 12.9 %) than in those with non-SNN neuropathy (11/301, 3.7 %). Among the 11 non-SNN neuropathy AGO1 Abs-positive cases, there were 3/80 (3.8 %) with SFN, suggesting a limited involvement of AGO1-Abs in this group of patients. However, the clinical features of these population were not detailed and further investigation in larger cohorts are warranted.

6 Post-COVID19 infection and vaccine SFN

COVID-19 infection and vaccination can be associated with peripheral nervous system manifestations, including Guillain-Barré syndrome and small fiber neuropathy (Gomez et al. 2023). A few studies reported the onset of SFN after COVID-19 infection. Burakgazi (2022) reported two cases of presumed SFN following COVID-19 infection, confirmed by a significantly reduced IEFND at the skin biopsy. Oaklander et al. (2022) analyzed a cohort of 17 patients with long COVID referred for peripheral neuropathy. Abnormal electrodiagnostic studies were shown in 16.7 % of cases, and SFN was confirmed by skin biopsies of the lower leg in 62.5 % of cases. In another study, including 16 patients with suspected post-COVID-19 SFN, nine were tested for SFN-related autoantibodies, and of these three were positive for TS-HDS and three for FGFR3 antibodies (McAlpine et al. 2024). Few small studies highlighted the possible correlation between SFN and SARS-CoV-2 vaccines (Mastropaolo and Hasbani 2023; Safavi et al. 2022; Schelke et al. 2022; Waheed et al. 2021). The presence of autoantibodies was investigated in few cases (Mastropaolo and Hasbani 2023; Schelke et al. 2022). One patient was positive for FGFR3 antibodies (Mastropaolo and Hasbani 2023) whereas other cases had high angiotensin converting enzyme 2 (ACE2) antibody titers (Schelke et al. 2022).

ACE2 is an enzyme expressed in a membrane-bound and soluble form responsible for converting vasoconstricting angiotensin II (Ang II) to angiotensin (1–7). Ang II induces immune activation by binding to the AT1 receptor, while Ang (1–7) binds to the Mas receptor to decrease inflammation (Shirbhate et al. 2021). It is known that the spike protein of the SARS-CoV-2 virus binds to ACE2, reducing its enzymatic activity and enhancing the immune response (Kuba et al. 2005). Interestingly, a study reported the expression of ACE2 in human DRG neurons, particularly nociceptors, suggesting a potential target for SARS-CoV-2-invasion of the sensory nervous system and evoking persistent symptoms (Shiers

et al. 2020). Moreover, the ACE2 could be the target of ACE2 autoantibodies, that can be detected in patients following COVID-19 infection (Arthur et al. 2021).

Some studies highlight the possible beneficial effects of corticosteroids, IVGV and PLEX treatment of SFN presumably induced by COVID-19 infection or vaccination (Mastropaolo and Hasbani 2023; McAlpine et al. 2024; Oaklander et al. 2022; Schelke et al. 2022). The efficacy of these treatments supports the postulated possibility of immune system involvement in the development of SFN after COVID-19 infection and vaccination; however, there is a need for more extended clinical studies to establish the usefulness of immunomodulatory treatments. Safavi et al. (2022) highlight the spontaneous improvement of a subgroup of patients that had not been treated with IVIG, outlying the need of further studies to verify the role of immunomodulatory treatment of SFN related to COVID-19. It would also be interesting to expand the analysis of the presence of ACE2 and other autoantibodies (possible biomarkers for SFN, reported in previous chapters) to have a better understanding of the autoimmune component in COVID-19 infection and vaccination related SFN cases.

7 Conclusion and future directions

Several evidence suggest that the immune system plays a role in neuropathic pain through autoimmune mechanisms associated with autoantibodies targeting nervous system antigens. However, proof of pathogenicity is still lacking for most of these antibodies and even in cases where pathogenicity has been proven, the mechanisms underlying pain and small fiber neuronal loss have not been entirely elucidated. Future studies should better define the frequency and associated clinical features of patients with autoimmune SFN, provide *in vitro* and *in vivo* studies and explore for new, still unidentified, autoimmune targets. Moreover, future studies should investigate their utility as possible biomarkers of response to immunotherapy.

Acknowledgments: Work supported by #NEXTGENERATIONEU (NGEU) and funded by the Ministry of University and Research (MUR), National Recovery and Resilience Plan (NRRP), project MNESYS (PE0000006) – A Multiscale integrated approach to the study of the nervous system in health and disease (DN. 1553 11.10.2022).

Research ethics: Not applicable.

Author contributions: LM writing original draft, data curation; LS data curation, visualization; FR data curation, writing (review and editing); IR data curation, writing (review and editing); VD, data curation, writing (review and

editing); MC, data curation, writing (review and editing); RL, supervision, writing (review and editing); MPG conceptualization, supervision, writing original draft. The author(s) have (has) accepted responsibility for the entire content of this manuscript and approved its submission.

Competing interests: The author(s) state(s) no conflict of interest.

Research funding: None declared.

Data availability: Not applicable.

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